

Installing R and RStudio

ECON 301

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This handout will briefly introduce how to install R and RStudio on Windows and OS X. Both of these will serve as our major tools throughout the semester to apply the econometric techniques that we learn in the lectures to real world data sets.

R is a language and environment for statistical computing and graphics. R Provides a wide variety of statistical (linear and nonlinear modeling, classical statistical tests, time-series analysis, classification, clustering, ...) and graphical techniques, and is highly extensible, meaning that you can extend its functionality by programming new routines and commands. In this course, we will use R to do most of our data analysis, to run linear regressions and perform statistical tests.

One of the major advantages of R is that it is free and open source software. This means that not only you pay nothing for it but also that its core mechanisms and routines are open to anyone to see. The other important aspect of R is that there are many routines and packages already available to use that implement new functionalities that you can download from CRAN, the *Comprehensive R Archive Network*.

On the other hand, RStudio is an integrated development environment (IDE) that is a powerful user interface for R and is also free. Of course, you can use R by itself, but RStudio makes life much easier.

1 Where to get the software

R and RStudio are separate pieces of software. **You will need to install R first.** Follow the next steps. Note that you will most likely need administrative privileges to perform both installs.

1. Download R from <http://cran.us.r-project.org/>
 - Windows: click on “Download R for Windows” → “base” → “Download R 4.x.x for Windows”
 - OS X click on “Download R for Mac OS X” → “R-4.x.x.pkg (latest version)”
2. Install R. Leave all default settings in the installation options.
3. Download RStudio from <http://rstudio.org/download/desktop>.
 - Windows: Just install it. Leave all default settings in the installation options.
 - OSX: Install RStudio by dragging the application icon to your Applications folder.

2 Documents for beginners and getting help

Usually, the internet is a good place to look for help. Most of the time you will be able to find an answer by just an appropriate search in Google.

For the times in which you need more structured advice, the following are good sources:

- R for Beginners by Emmanuel Paradis,
https://cran.r-project.org/doc/contrib/Paradis-rdebuts_en.pdf
- A (very) short introduction to R, by Paul Torfs & Claudia Brauer,
<https://cran.r-project.org/doc/contrib/Torfs+Brauer-Short-R-Intro.pdf>